

Surveillance Guidance Using Lamé Curve Paths

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This paper introduces a new approach for efficient rectangular boundary surveillance guidance for unmanned aerial vehicles (UAVs) using a continuous-curvature path described by a Lamé curve. Curvature and path length characteristics of the Lamé curve paths are analyzed, and conditions are deduced for the Lamé-curve-based circumscription of rectangular boundaries. Given the maximum turn rate capability of the UAV, a comparative study highlights significant reduction in path length as compared to the widely used elliptic circumscription. In conjunction with the desired Lamé curve path, a vector field guidance method is proposed for path-following. The vector field around the path is generated using a commanded heading angle, which encapsulates a tangent vector along the path and an arcsine shaping function of the UAV position error with respect to the desired path. Asymptotically converging behavior of the UAV position error is ascertained, and simulation studies illustrate the accurate following of the desired path for arbitrary UAV initial conditions. Indoor experimental studies utilizing a quadrotor UAV platform present the practical applicability of the proposed surveillance guidance solution.

Nomenclature

A, B, C, D, O	= vertices and center of rectangular boundary, respectively
a, b, n	= semimajor, semiminor axis, and Lamé curve parameter, respectively
F	= implicit function representing a Lamé curve
g	= acceleration due to gravity, $\text{m} \cdot \text{s}^{-2}$
k, r	= order of differentiation and order of differentiability, respectively
k_e	= guidance gain
L, S	= one-fourth of the perimeter and the perimeter of Lamé curve, respectively, m
$(l, m), \mu$	= length (m), breadth (m), and aspect ratio of rectangular boundary, respectively
M	= mass of Crazyflie 2.0, kg
P	= a point on the desired Lamé curve path
Q	= instantaneous UAV position
$\mathbb{R}$	= set of real numbers
U	= set of points satisfying Lamé curve
u	= input vector
u_1	= total thrust, N
X, Y	= axes of a Cartesian frame
x, y, z	= cartesian coordinates
$\dot{x}_{\text{ref}}, \dot{y}_{\text{ref}}, \dot{z}_{\text{ref}}$	= reference velocity commands for experimental flight, $\text{m} \cdot \text{s}^{-1}$
$\dot{x}_u, \dot{y}_u$	= instantaneous UAV speed components, $\text{m} \cdot \text{s}^{-1}$
$\ddot{x}_c, \ddot{y}_c, \ddot{z}_c$	= commanded translational acceleration for experimental flight, $\text{m} \cdot \text{s}^{-2}$
α	= square root of sum of squares of F_x^f and F_y^f
β	= path-tracking parameter

θ	= angular position in polar coordinates
κ, κ_u	= curvature of Lamé curve and curvature of instantaneous UAV path, respectively, m^{-1}
ζ	= Lamé curve as a zero level set
$\phi, \pm \phi_{\max}$	= UAV bank angle and maximum bank angle limit, respectively, rad
$\psi, \dot{\psi}$	= UAV heading angle (rad), and turn rate ($\text{rad} \cdot \text{s}^{-1}$), respectively
$\psi(0), \dot{\psi}(0)$	= initial heading angle (rad), and initial turn rate ($\text{rad} \cdot \text{s}^{-1}$), respectively
ψ_t, ψ_o	= path-tangential term and shaping function, respectively, rad
$\psi_q, \dot{\psi}_q$	= crazyflie yaw angle (rad) and yaw rate ($\text{rad} \cdot \text{s}^{-1}$), respectively
$\dot{\psi}_t, \dot{\psi}_o$	= path-tangential rate and shaping function rate, respectively, $\text{rad} \cdot \text{s}^{-1}$

Subscripts

$c, d, u, 0$	= commanded, desired, instantaneous, and initial value, respectively
g	= with respect to ground
p	= point on the curve
$(p_x, p_y, p_z), (dx, dy, dz)$	= proportional- and derivative-gain components, respectively
ref, max	= reference and maximum value
$(x, y), (xx, yy)$	= first- and second-order partial derivatives with respect to x and y , respectively

Superscripts

f	= family of curves
$*$	= optimal value
$+$	= positive number

I. Introduction

ENCOMPASSING missions such as patrolling a convoy of vehicles, safeguarding installation, and monitoring facilities, aerial surveillance using unmanned aerial vehicles (UAVs) is increasing in demand. These missions often require accurate tracking of enclosed rectangular area boundaries. Sharp corners in a rectangular boundary prohibit accurate path-following using turn-rate-constrained fixed-wing UAVs. Dubins paths can be used for circumscribing closed boundaries while abiding by a given UAV maximum turn rate [1]. However, the straight-line–arc junctions in Dubins paths present discontinuities in curvature, which make them

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dynamically infeasible to be accurately tracked by a fixed-wing UAV. Consequently, approximating rectangular boundaries with a smooth and continuous curvature path provides a feasible boundary surveillance solution. This work is motivated by the potential use of a family of Lamé curve paths in describing rectangular area boundaries with more versatility and flexibility vs widely used circular and elliptical approximations.

Enclosed area boundaries are often circumscribed using circular [2,3] or elliptic curves [4–6] to generate feasible surveillance paths. Reference [2] addresses an area coverage problem, wherein the coverage area is divided into enclosed square boundaries, and a UAV loiters around the circumscribing circular paths for persistent surveillance. Another rectangular area surveillance problem is considered in Ref. [3], where the UAVs follow paths with turn-rate and field-of-view constraints. Refs. [4,5] developed strategies for bounding a moving convoy within a rectangular boundary and monitoring it using an elliptic path circumscribing the boundary. To follow the circumscribing elliptic path, a vector-field heading guidance command is used. In a subsequent work [6], speed control for each UAV is proposed to achieve equiangular phase separation in multi-UAV cooperative surveillance over elliptic paths. Circular circumscription of a rectangular boundary is inefficient because of the high path length of the circumscribing path. In contrast, elliptic circumscription offers better path-length characteristics. However, elliptic circumscription may also present a path length that is significantly higher than the perimeter of the rectangular boundary. For a better circumscription of rectangular boundaries, this work proposes Lamé curve as an alternative smooth circumscribing path. Lamé curve was introduced by French mathematician Gabriel Lamé [7] in 1818. Danish architect Piet Hein further explored Lamé curve for a variety of purposes, including motorway bridges and other architectural applications [8]. References [9–12] use Lamé curves as reference paths for UAVs. However, these works do not investigate the utility of Lamé curve paths in a path-planning scenario. Further, UAVs need guidance logic to autonomously follow any prescribed path. In that regard, this work also develops a vector-field guidance logic for following the proposed Lamé curve surveillance paths.

Vector-field-based algorithms present an elegant guidance approach for path-following applications. The vector-field approach utilizes a position-based guidance command that ensures a unique heading direction at each point in the domain, effectively guiding the UAV to the desired path. As a result, the guidance input in the vector-field method is resilient to position disturbances and the effects of vehicle dynamics. These algorithms also present a promising solution for achieving low path-following errors and minimal control effort, while utilizing a simple and easily computable guidance logic [13]. Vector-field guidance methods can be categorized into two methods, namely, gradient-based [14–21] and direct methods [22–25], and are briefly reviewed subsequently.

Gradient-based methods derive the commanded heading direction from the gradient of a potential function designed to reach its minimum along the desired path. Conversely, direct methods determine the desired heading direction directly from the UAV's current position relative to the path. Reference [14] introduced a Lyapunov vector-field method, where the gradient of a potential function defines an attracting limit cycle for a specified circular path. In this method, the gradient is used to derive the heading guidance command, which includes a contraction term that governs the radial speed of the UAV and a circulation term that regulates the tangential speed relative to the desired path. By proposing the circulation parameter as a function of the radial distance, Ref. [15] presented a vector-field method that provides faster settling onto the desired path with a lower maximum curvature, as compared to the method presented in Ref. [14]. As further extensions, the Lyapunov vector-field method is developed for following a dynamic star curve in Ref. [16] and a variety of planar curves in Ref. [12]. References [17,18] employ the concept of calculating the desired heading direction from the vector sum of tangential and normal components relative to the desired path. These two methodologies differ in how they construct their tangential components within the vector field. Reference [17] derives the tangential component as the cross product of the gradients of zero level sets of functions, whose intersection describes the desired path. Conversely, Ref. [18] directly

determines the tangential component by rotating the normal vector, which is derived from the gradient of an implicit representation of the desired path. Reference [19] uses a repulsive vector field in combination with the path-following method used in Ref. [17] for avoiding circular obstacles. Although the associated computations can be simplified as discussed in Refs. [20,21], gradient-based vector-field methods necessitate the computation of gradients to determine the guidance command.

In contrast to gradient-based methods, Refs. [22–25] present a class of vector fields where the UAV heading direction can be uniquely and directly determined using the UAV's position relative to the desired path. As a result, direct methods are computationally efficient, simple, and easily implementable forms of vector-field guidance logic. Using an inverse tangent function of the UAV position error, Ref. [22] proposes a vector-field guidance method for following straight-line and circular paths. By using the tangent information along the desired path, Ref. [23] further develops the guidance law used in Ref. [22] to follow a sinusoidal path. Using an arcsine function, Ref. [24] presents a direct vector-field guidance law for following straight-line and circular paths while using lower maximum UAV turn rate, as compared to Ref. [22]. Reference [25] presents some preliminary results on arcsine-function-based vector-field guidance method for following elliptic paths. However, Refs. [22–25] do not consider following Lamé curve paths.

The major contribution of this work is threefold:

1) Circumscribing a rectangular boundary using a continuous- and bounded-curvature Lamé curve path is proposed. Geometrical properties, such as curvature and path length of Lamé curve paths are analyzed, and analytical conditions for rectangular boundary circumscription using a family of Lamé curve paths are deduced. By considering the maximum turn-rate capability of the UAV, a comparative study shows that Lamé curve significantly reduces the circumscribing path length as compared to the widely used elliptic circumscription.

2) A direct vector-field method utilizing heading angle guidance command is introduced to follow the proposed Lamé curve path. The guidance command comprises a path-tangential term and a shaping function wherein the path-tangential term governs the UAV speed along the path and the shaping function controls the UAV speed of convergence toward the desired path. The asymptotically converging behavior of the UAV position error is established using Lyapunov's direct method. The proposed vector-field method presents a simple, analytically tractable and easily computable guidance solution. Further, as per the best of the authors' knowledge, this is the first of its kind, direct vector-field method that is specifically and fully developed for following Lamé curve paths.

3) Realistic validation studies, including simulation and experimental results, are presented to verify the proposed guidance method. Simulation studies considering a second-order heading-hold autopilot with several UAV initial conditions demonstrate accurate path-following from arbitrary initial positions. Indoor experimental studies using a quadrotor UAV platform and a motion capture facility demonstrate the practical applicability of the proposed surveillance guidance solution.

The remainder of this paper is organized as follows: Sec. II discusses basic properties of Lamé curves. The surveillance problem is formulated in Sec. III. Circumscribing a rectangular boundary using Lamé curve is discussed in Sec. IV. Section V discusses the proposed vector-field guidance law, and simulation studies are discussed in Sec. VI. Experimental validation results are presented in Sec. VII. Section VIII provides concluding remarks.

II. Lamé Curve: Basic Properties

Lamé curve is a two-dimensional symmetric closed curve described by the three-parameter $(a, b, n \in \mathbb{R}^+)$ geometric equation [26],

$$\left| \frac{x}{a} \right|^n + \left| \frac{y}{b} \right|^n = 1 \quad (1)$$

where a, b represent the semimajor and semiminor axes, and the parameter n governs the shape and convexity of the curve. For $n = 2$, Eq. (1) describes the standard ellipse.

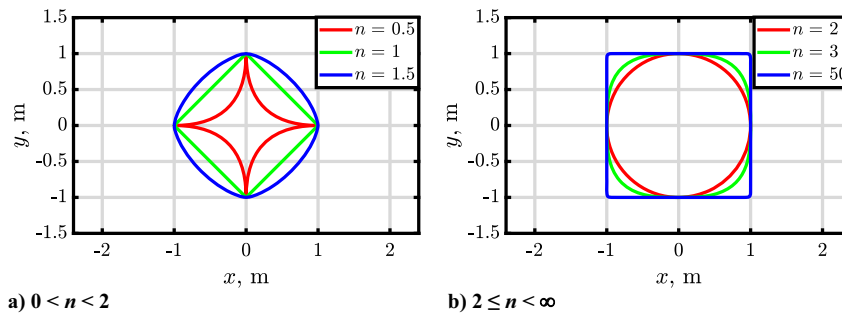


Fig. 1 Family of Lamé curves for $a = b = 1$.

Considering $a = b = 1$, Fig. 1 represents a family of Lamé curves for different values of n . From the figure, it can be seen that the shape and slope continuity of the Lamé curve varies significantly with the choice of n .

Equation (1) can be expressed in an implicit form as

$$F(x, y) = \left| \frac{x}{a} \right|^n + \left| \frac{y}{b} \right|^n - 1 = 0 \quad (2)$$

Remark 1: Changing variables x to $-x$ and y to $-y$ in Eq. (2) does not change $F(x, y)$, which proves the symmetricity of the Lamé curve with respect to x and y axes.

A. Differentiability of Lamé Curves

Definition 1: A function $F: U \rightarrow \mathbb{R}$, $U \subset \mathbb{R}^2$ is said to be continuously differentiable of order r , if partial derivatives $\partial^k F / \partial x^k$, $\partial^k F / \partial y^k$, for $k = 1, 2, \dots, r$ of the function exist and are continuous for all points $(x, y) \in U$.

Proposition 1: For $0 < n \leq 1$, the Lamé curve is differentiable everywhere except at the points $(x, y) \in \{(0, b), (0, -b), (a, 0), (-a, 0)\}$, and for $n > 1$ it is differentiable everywhere.

Proof: In Eq. (2), the function $F: \mathbb{R}^2 \rightarrow \mathbb{R}$ is the set of ordered pairs $X \times Y = \{(x, y) : x \in X, y \in Y\}$ where sets $X = \{x \in \mathbb{R} : -a \leq x \leq a\}$ and $Y = \{y \in \mathbb{R} : -b \leq y \leq b\}$. Equation (2) implies

$$F(x, y) = \begin{cases} \frac{x^n}{a^n} + \frac{y^n}{b^n} = 1, & x \in [0, a], \quad y \in (0, b] \\ \frac{(-x)^n}{a^n} + \frac{y^n}{b^n} = 1, & x \in [-a, 0), \quad y \in [0, b) \\ \frac{(-x)^n}{a^n} + \frac{(-y)^n}{b^n} = 1, & x \in (-a, 0], \quad y \in [-b, 0) \\ \frac{x^n}{a^n} + \frac{(-y)^n}{b^n} = 1, & x \in (0, a], \quad y \in (-b, 0] \end{cases} \quad (3)$$

It can be readily seen that $F(x, y)$ in Eq. (3) is differentiable for all $\{(x, y) : 0 < x < a, 0 < y < b\}$. Similarly, Eq. (3) implies that $F(x, y)$ is also differentiable for all points in the set $\{(x, y) : -a < x < 0, 0 < y < b\} \cup \{(x, y) : -a < x < 0, -b < y < 0\} \cup \{(x, y) : 0 < x < a, -b < y < 0\}$. Differentiability of $F(x, y)$ at points $(0, b)$, $(0, -b)$, $(a, 0)$ and $(-a, 0)$ is analyzed subsequently. Consider the point $(x, y) = (0, b)$. Evaluating $(\partial F / \partial x)|_{x=0^+}$ using Eq. (3),

$$\left. \frac{\partial F}{\partial x} \right|_{x=0^+} = \lim_{x \rightarrow 0^+} \frac{nx^{n-1}}{a^n} \begin{cases} \rightarrow +\infty, & 0 < n < 1 \\ = \frac{1}{a}, & n = 1 \\ = 0, & n > 1 \end{cases} \quad (4)$$

Computing $(\partial F / \partial x)|_{x=0^-}$ using Eq. (3),

$$\left. \frac{\partial F}{\partial x} \right|_{x=0^-} = \lim_{x \rightarrow 0^-} \frac{-n(-x)^{n-1}}{a^n} \begin{cases} \rightarrow -\infty, & 0 < n < 1 \\ = -\frac{1}{a}, & n = 1 \\ = 0, & n > 1 \end{cases} \quad (5)$$

Further, considering point $(x, y) = (a, 0)$ and evaluating $(\partial F / \partial y)|_{y=0^+}$ and $(\partial F / \partial y)|_{y=0^-}$ using Eq. (3), respectively,

$$\left. \frac{\partial F}{\partial y} \right|_{y=0^+} = \lim_{y \rightarrow 0^+} \frac{ny^{n-1}}{b^n} \begin{cases} \rightarrow +\infty, & 0 < n < 1 \\ = +\frac{1}{b}, & n = 1 \\ = 0, & n > 1 \end{cases} \quad (6)$$

$$\left. \frac{\partial F}{\partial y} \right|_{y=0^-} = \lim_{y \rightarrow 0^-} -\frac{ny^{n-1}}{b^n} \begin{cases} \rightarrow -\infty, & 0 < n < 1 \\ = -\frac{1}{b}, & n = 1 \\ = 0, & n > 1 \end{cases} \quad (7)$$

Case 1: $0 < n \leq 1$

From Eqs. (4) and (5) for $0 < n \leq 1$, $(\partial F / \partial x)|_{x=0^+} \neq (\partial F / \partial x)|_{x=0^-}$, which implies that the Lamé curve is not differentiable at $(0, b)$. Further, using the symmetricity property discussed in Remark 1, the Lamé curve is also not differentiable at $(0, -b)$. Similarly, Eqs. (6) and (7) and Remark 1 imply that Lamé curve is not differentiable at $(a, 0)$ and $(-a, 0)$.

Case 2: $n > 1$

For $n > 1$, Eqs. (4) and (5) imply $(\partial F / \partial x)|_{x=0^+} = (\partial F / \partial x)|_{x=0^-}$. Further, using Eq. (3), $(\partial F / \partial y)|_{y=b}$ exists, which proves that the Lamé curve is differentiable at $(0, b)$, and using Remark 1, it is differentiable also at $(0, -b)$. From Eqs. (6) and (7) for $n > 1$, $(\partial F / \partial y)|_{y=0^+} = (\partial F / \partial y)|_{y=0^-}$. Further, using Eq. (3), $(\partial F / \partial x)|_{x=a}$ exists, hence the Lamé curve is differentiable at $(a, 0)$, which, using Remark 1, also proves that it is differentiable at $(-a, 0)$. $\square$

B. Curvature Characteristics of Lamé Curves

As a potential UAV path, curvature characteristics of Lamé curve is of importance. Consider the implicit form of Lamé curve [Eq. (2)]; its curvature κ can be expressed as [27]

$$\kappa = \frac{|F_y^2 F_{xx} - 2F_x F_y F_{xy} + F_x^2 F_{yy}|}{(F_x^2 + F_y^2)^{3/2}} \quad (8)$$

where $F_x = \partial F / \partial x$, $F_y = \partial F / \partial y$, $F_{xx} = \partial^2 F / \partial x^2$, $F_{xy} = \partial^2 F / \partial x \partial y$, and $F_{yy} = \partial^2 F / \partial y^2$, respectively.

Proposition 2: For $1 < n < 2$, Lamé curve has finite curvature everywhere except at points $(x, y) \in \{(0, b), (0, -b), (a, 0), (-a, 0)\}$, and for $2 \leq n < \infty$ it has finite curvature everywhere.

Proof: Using Eq. (3),

$$F_x = \frac{nx^{n-1}}{a^n} \quad (9)$$

$$F_{xx} = \frac{n(n-1)x^{n-2}}{a^n} \quad (10)$$

$$F_y = \frac{ny^{n-1}}{b^n} \quad (11)$$

$$F_{yy} = \frac{n(n-1)y^{n-2}}{b^n} \quad (12)$$

$$F_{xy} = 0 \quad (13)$$

Using Eqs. (9)–(13) in Eq. (8),

$$\kappa = \frac{(n-1)a^{2n}b^{2n}x^{n-2}y^{n-2}}{(b^{2n}x^{2(n-1)} + a^{2n}y^{2(n-1)})^{3/2}}, \quad x \in [0, a], \quad y \in (0, b] \quad (14)$$

Similarly, computing partial derivatives using Eq. (3) and using them in Eq. (8),

$$\kappa = \begin{cases} \frac{(n-1)a^{2n}b^{2n}(-1)^{n-2}x^{n-2}y^{n-2}}{(b^{2n}x^{2(n-1)} + a^{2n}y^{2(n-1)})^{3/2}}, & x \in [-a, 0), \quad y \in [0, b) \\ \frac{(n-1)a^{2n}b^{2n}x^{n-2}y^{n-2}}{(b^{2n}x^{2(n-1)} + a^{2n}y^{2(n-1)})^{3/2}}, & x \in (-a, 0], \quad y \in [-b, 0) \\ \frac{(n-1)a^{2n}b^{2n}(-1)^{n-2}x^{n-2}y^{n-2}}{(b^{2n}x^{2(n-1)} + a^{2n}y^{2(n-1)})^{3/2}}, & x \in (0, a], \quad y \in (-b, 0] \end{cases} \quad (15)$$

For $n > 1$, it can be readily seen that Eqs. (14) and (15) yield finite curvature for all $\{(x, y) : 0 < x < a, 0 < y < b\} \cup \{(x, y) : -a < x < 0, 0 < y < b\} \cup \{(x, y) : -a < x < 0, -b < y < 0\} \cup \{(x, y) : 0 < x < a, -b < y < 0\}$. Further, curvature at points $(0, b)$, $(0, -b)$, $(a, 0)$ and $(-a, 0)$ is discussed subsequently.

Case 1: $1 < n < 2$

Equation (14) implies $\kappa \rightarrow \infty$ at point $(x, y) = (0, b)$. Using Remark 1, the curvature is hence infinite also at $(0, -b)$. Similarly, Eq. (15) and Remark 1 imply that for $1 < n < 2$ the Lamé curve has infinite curvature at $(a, 0)$ and $(-a, 0)$.

Case 2: $n \geq 2$

Equations (14) and (15) imply $\kappa \rightarrow 0$ at points $(x, y) \in \{(0, b), (-a, 0), (0, -b), (a, 0)\}$. $\square$

The curvature of Lamé curve can also be expressed using parametric form, which is given by

$$x = a|\cos \theta|^{2/n} \text{sign}(\cos \theta), \quad y = b|\sin \theta|^{2/n} \text{sign}(\sin \theta) \quad (16)$$

where parameter $\theta \in [0, 2\pi)$ is measured in an anticlockwise direction from the positive x axis. For $\theta \in [0, \pi/2]$, Eq. (16) reduces to

$$x = a \cos^{2/n} \theta, \quad y = b \sin^{2/n} \theta \quad (17)$$

which, on substituting in Eq. (14), leads to

$$\kappa = \frac{(n-1)ab \cos^{\frac{2(n-2)}{n}} \theta \sin^{\frac{2(n-2)}{n}} \theta}{(b^2 \cos^{\frac{4(n-1)}{n}} \theta + a^2 \sin^{\frac{4(n-1)}{n}} \theta)^{3/2}}, \quad \theta \in [0, \pi/2] \quad (18)$$

Taking into account Propositions 1 and 2, we consider

$$2 \leq n < \infty \quad (19)$$

as the feasible range of n for generating continuous and bounded-curvature Lamé curve paths.

Remark 2: For $n > 2$, Eq. (18) cannot be solved analytically to deduce extremum curvature. Reference [28] presents some approximations for maximum curvature, as n tends to a very large value.

C. Path Length of Lamé Curves

For $\theta \in [0, \pi/2]$, the Lamé-curve arc length L can be expressed as

$$L = \int_0^{\pi/2} \sqrt{\left(\frac{\partial x}{\partial \theta}\right)^2 + \left(\frac{\partial y}{\partial \theta}\right)^2} d\theta \quad (20)$$

where $\partial x/\partial \theta$, $\partial y/\partial \theta$ are partial derivatives that can be deduced using Eq. (17) as

$$\frac{\partial x}{\partial \theta} = -\frac{2a}{n} \cos^{\frac{2-n}{n}} \theta \sin \theta, \quad \frac{\partial y}{\partial \theta} = \frac{2b}{n} \sin^{\frac{2-n}{n}} \theta \cos \theta \quad (21)$$

Using Eqs. (20) and (21),

$$L = \int_0^{\pi/2} \sqrt{\frac{4a^2}{n^2} \cos^{\frac{2(2-n)}{n}} \theta \sin^2 \theta + \frac{4b^2}{n^2} \sin^{\frac{2(2-n)}{n}} \theta \cos^2 \theta} d\theta \quad (22)$$

Further, using Eq. (22) and Remark 1, path-length S of a complete Lamé curve can be expressed as

$$S = 4L = 4 \int_0^{\pi/2} \sqrt{\frac{4a^2}{n^2} \cos^{\frac{2(2-n)}{n}} \theta \sin^2 \theta + \frac{4b^2}{n^2} \sin^{\frac{2(2-n)}{n}} \theta \cos^2 \theta} d\theta \quad (23)$$

Remark 3: For $n \geq 2$, Eq. (23) does not offer an analytic solution.

III. Problem Formulation

Consider a planar rectangular boundary $ABCD$ as shown in Fig. 2. The boundary is centered at origin O , and has length $AD = BC = l$ and width $AB = DC = m$, where $l > m > 0$. Given the curvature discontinuity at the corners of the boundary, accurate boundary tracking using a fixed-wing UAV is not possible due to its limited turning-rate capability. Consequently, the objectives of this work are twofold:

1) First, to design a prospective circumscribing path (as indicated in Fig. 2) with desirable curvature and path-length characteristics.

2) Second, to develop a vector field for guiding the UAV from an arbitrary initial position onto the circumscribing path.

The kinematic equations of motion of the UAV can be described by the following equations:

$$\dot{x} = V_a \cos \psi \quad (24)$$

$$\dot{y} = V_a \sin \psi \quad (25)$$

where (x, y) , V_a , and ψ represent UAV position in a fixed frame, airspeed, and heading angle, respectively.

IV. Lamé Curve Circumscribing a Rectangular Boundary

Consider a rectangular boundary $ABCD$ with vertices $A[-(l/2), (m/2)]$, $B[-(l/2), -(m/2)]$, $C[(l/2), -(m/2)]$, and $D[(l/2), (m/2)]$ to be circumscribed by a Lamé curve as shown in Fig. 3.

Proposition 3: The parameters (a, b, n) of a family of Lamé curves circumscribing a rectangle with vertices $A[-(l/2), (m/2)]$, $B[-(l/2), -(m/2)]$, $C[(l/2), -(m/2)]$, and $D[(l/2), (m/2)]$ satisfy

$$\left(\frac{l}{2a}\right)^n + \left(\frac{m}{2b}\right)^n = 1 \quad (26)$$

where $l/2 < a < \infty$, and $m/2 < b < \infty$.

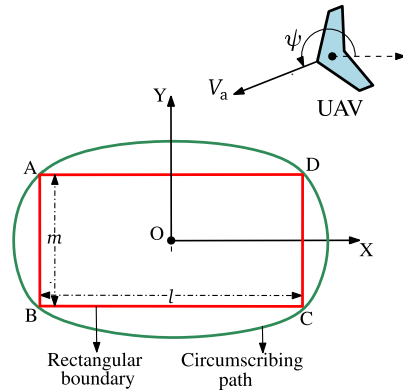


Fig. 2 Problem scenario.

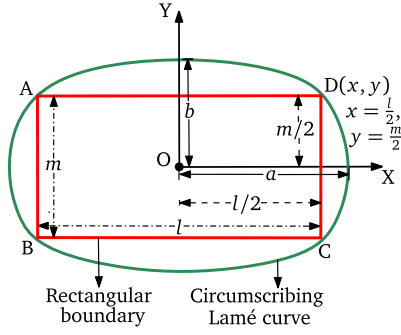


Fig. 3 Circumscribing Lamé curve.

Proof: Consider that the Lamé curve described by Eq. (1) passes through the vertex $D[(l/2), (m/2)]$. That, using Remark 1, implies that the Lamé curve also passes through the other three vertices. Using $(x, y) = [(l/2), (m/2)]$ in Eq. (1),

$$\left(\frac{l}{2a}\right)^n + \left(\frac{m}{2b}\right)^n = 1 \quad (27)$$

Using Eq. (27), a and b can be expressed, respectively, as

$$a = \frac{bl}{(2^n b^n - m^n)^{1/n}} \quad (28)$$

$$b = \frac{am}{(2^n a^n - l^n)^{1/n}} \quad (29)$$

Using Eqs. (28) and (29), for finite real solutions of (a, b) , the following conditions need to be satisfied:

$$0 < 2^n b^n - m^n < \infty \implies m/2 < b < \infty \quad (30)$$

$$0 < 2^n a^n - l^n < \infty \implies l/2 < a < \infty \quad (31)$$

□

Substituting $n = 2$ in Eq. (26) leads to the special case of a family of ellipses circumscribing the rectangular boundary. An algorithm for generating minimum-length Lamé curve path circumscribing a rectangular boundary while abiding by a maximum curvature bound is discussed next.

Because the problem scenario involves planar fixed-wing UAV motion, the heading-rate constraint associated with a coordinated turn is considered. For a coordinated turn, the corresponding bank angle ϕ demand can be expressed as [29]

$$\tan \phi = \frac{V_a}{g} \dot{\psi} \quad (32)$$

$$\implies \dot{\psi} = \frac{g \tan \phi}{V_a} \quad (33)$$

Also, the curvature κ_u of the instantaneous UAV path and heading-rate $\dot{\psi}$ are related through airspeed of the UAV V_a as

$$\kappa_u = \frac{\dot{\psi}}{V_a} \quad (34)$$

Considering the UAV bank angle constraint $\phi_{\max}$, speed V_a , and acceleration due to gravity g while using Eqs. (33) and (34), the corresponding maximum turn-rate $\dot{\psi}_{\max}$ and maximum curvature $\kappa_{\max_u}$, respectively, are obtained as

$$\dot{\psi}_{\max} = \frac{g \tan \phi_{\max}}{V_a}; \quad \kappa_{\max_u} = \frac{\dot{\psi}_{\max}}{V_a} \quad (35)$$

Flight endurance plays a crucial role in accomplishing mission objectives, and it primarily depends on the path length traversed by the UAV. Therefore, using Eq. (23), the Lamé curve with the least path-length S is computed while satisfying maximum UAV curvature limits, as explicated in Algorithm 1. Therein, $\kappa_{\max}$ represents the maximum curvature of the Lamé curve path under consideration.

Algorithm 1: Lamé curve circumscribing rectangular boundary

- 1: Consider the UAV curvature limit $\kappa_{\max_u}$, speed V_a , and dimensions (l, m) of the rectangular boundary.
- 2: Deduce all possible Lamé curves with $\kappa_{\max} = \kappa_{\max_u}$ while satisfying Eqs. (19), (26), (28), and (29) and compute path-length S for each of the Lamé curves, using Eq. (23).
- 3: From the set of Lamé curves deduced in Step 2, determine the Lamé curve with the least path length characterized by optimal path parameters (n^*, a^*, b^*) .

V. Proposed Guidance Law

This section discusses the vector-field guidance design for following the Lamé curve paths introduced in Sec. IV.

Using Eq. (2), a Lamé curve can be defined as the zero set of a function $F(x, y)$ for all $(x, y) \in \mathbb{R}^2$, expressed as

$$\zeta \triangleq \{(x, y) : F(x, y) = (\beta - 1) = 0\} \subset \mathbb{R}^2 \quad (36)$$

where β is a dimensionless parameter defined as

$$\beta \triangleq \left|\frac{x}{a}\right|^n + \left|\frac{y}{b}\right|^n \quad (37)$$

Consequently, $\beta > 1$ and $\beta < 1$ imply $F(x, y) > 0$ and $F(x, y) < 0$, which corresponds to all points lying outside and inside the desired Lamé curve ($\beta = 1$), respectively.

A. Guidance Law

To guide the UAV from an arbitrary initial position onto the desired Lamé curve path, the commanded heading angle ψ_d is proposed as

$$\psi_d = \begin{cases} \psi_t - \psi_o, & 0 < \beta \leq 1 \quad (\text{Inside}) \\ \psi_t + \psi_o, & \beta \geq 1 \quad (\text{Outside}) \end{cases} \quad (38)$$

where path-tangential term ψ_t governs the UAV motion along the Lamé curve, and shaping function ψ_o regulates the rate of convergence toward the curve. The path-tangential term ψ_t at instantaneous UAV position $Q(x_u, y_u) \in \mathbb{R}^2$ can be determined as the slope at that point of a concentric Lamé curve from the family

$$\left|\frac{x}{a}\right|^n + \left|\frac{y}{b}\right|^n = \beta \quad (39)$$

where

$$\beta \triangleq \left|\frac{x_u}{a}\right|^n + \left|\frac{y_u}{b}\right|^n \quad (40)$$

This is illustrated in Fig. 4.

Lemma 1: As depicted in Fig. 4, consider $\tan \psi_t$ to be the slope at the point $Q(x_u, y_u)$ of the Lamé curve described by Eqs. (39) and (40). Further, point P represents the intersection of the desired Lamé curve and the line joining the origin to the point Q . On the desired Lamé curve, the slope at P is equal to $\tan \psi_t$.

Proof: See the Appendix. □

From Eq. (39), a family of concentric Lamé curve $F^f(x, y)$ can be expressed as

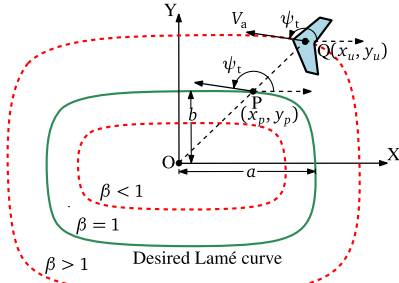


Fig. 4 Logic for computing ψ_t .

$$F^f(x, y) = \left| \frac{x}{a\beta^{1/n}} \right|^n + \left| \frac{y}{b\beta^{1/n}} \right|^n - 1 = 0 \quad (41)$$

where $\beta \in \mathbb{R}^+$ and $\beta \neq 1$. Further, Eq. (41) implies

$$F^f(x, y) = \begin{cases} \frac{x^n}{a^n\beta} + \frac{y^n}{b^n\beta} = 1, & x \in [0, a\beta^{1/n}), y \in (0, b\beta^{1/n}] \\ \frac{(-x)^n}{a^n\beta} + \frac{y^n}{b^n\beta} = 1, & x \in [-a\beta^{1/n}, 0), y \in [0, b\beta^{1/n}] \\ \frac{(-x)^n}{a^n\beta} + \frac{(-y)^n}{b^n\beta} = 1, & x \in (-a\beta^{1/n}, 0], y \in [-b\beta^{1/n}, 0) \\ \frac{x^n}{a^n\beta} + \frac{(-y)^n}{b^n\beta} = 1, & x \in (0, a\beta^{1/n}], y \in (-b\beta^{1/n}, 0] \end{cases} \quad (42)$$

Accordingly, using Eq. (42), the path-tangential term can be deduced for all quadrants as

$$\psi_t = \tan^{-1} \left(\frac{dy}{dx} \right) \bigg|_{(x,y)} = \tan^{-1} \left(\frac{F_x^f}{-F_y^f} \right), x \in [-a\beta^{1/n}, a\beta^{1/n}], y \in [-b\beta^{1/n}, b\beta^{1/n}] \quad (43)$$

$$F_x^f = \frac{\partial F^f(x, y)}{\partial x} = \begin{cases} \frac{nx^{n-1}}{a^n}, & x \in (0, a\beta^{1/n}] \\ \frac{-n(-x)^{n-1}}{a^n}, & x \in [-a\beta^{1/n}, 0) \end{cases} \quad (44)$$

and

$$F_y^f = \frac{\partial F^f(x, y)}{\partial y} = \begin{cases} \frac{ny^{n-1}}{b^n}, & y \in (0, b\beta^{1/n}] \\ \frac{-n(-y)^{n-1}}{b^n}, & y \in [-b\beta^{1/n}, 0) \end{cases} \quad (45)$$

The shaping function ψ_o in Eq. (38) is proposed as

$$\psi_o = \frac{\pi}{2} - \sin^{-1} \left(\frac{1}{1 + k_e(\beta - 1)^2} \right) \quad (46)$$

where $k_e > 0$ denotes the guidance constant and $(\beta - 1)$ represents the corresponding tracking error. It can be readily seen from Eq. (46) that as $\beta \rightarrow \infty$, $\psi_o \rightarrow \pi/2$, which upon substitution in Eq. (38), leads to $\psi_d \rightarrow \psi_t + \pi/2$. Further, with a steady decrease in β , the argument of arcsine function increases, which in turn decreases ψ_o and finally reduces it to zero on the desired Lamé curve path with $\beta = 1$. Consequently, the UAV heading angle complies with $\psi_d = \psi_t$, which is the desired direction of motion along the Lamé curve path. Similarly, for $0 < \beta < 1$, the UAV heading angle varies from $\psi_d \rightarrow \psi_t - \pi/2 + \sin^{-1}(1/(1 + k_e))$ for $\beta \rightarrow 0$ to $\psi_d = \psi_t$ at $\beta = 1$ with steady variation in the argument $(1/(1 + k_e(\beta - 1)^2))$.

B. Stability Analysis

The asymptotic behavior of the proposed vector-field guidance method's path-following error is verified in Proposition 4.

Proposition 4: Subject to the UAV kinematic model of Eqs. (24) and (25) and the heading angle governed by Eq. (38), the tracking error $(\beta - 1)$ converges asymptotically to zero for any initial $\beta_0 \in (0, \infty) \setminus \{1\}$.

Proof: Consider a Lyapunov function $W(\beta) = (\beta - 1)^2/2$, such that $W(\beta) = 0$ at $\beta = 1$ and $W(\beta) > 0$ for all $\beta \in (0, \infty) \setminus \{1\}$. Differentiating W with respect to time,

$$\dot{W} = (\beta - 1)\dot{\beta} \quad (47)$$

Using Eq. (39), for all points $(x, y) \in \mathbb{R}^2$,

$$\dot{\beta} = F_x^f \dot{x} + F_y^f \dot{y} \quad (48)$$

Using Eqs. (24), (25), (38), (47), and (48),

$$\begin{aligned} \dot{W} &= (\beta - 1)(V_a F_x^f \cos \psi_d + V_a F_y^f \sin \psi_d) \\ \Rightarrow \dot{W} &= \alpha(\beta - 1)V_a \left(\frac{F_x^f}{\alpha} \cos \psi_d + \frac{F_y^f}{\alpha} \sin \psi_d \right) \end{aligned} \quad (49)$$

where

$$\alpha \triangleq \sqrt{(F_x^f)^2 + (F_y^f)^2} \quad (50)$$

Further, using Eqs. (43–45),

$$\sin \psi_t = \frac{F_x^f}{\alpha} \quad \text{and} \quad \cos \psi_t = -\frac{F_y^f}{\alpha} \quad (51)$$

Finally, using Eqs. (49–51) and rearranging,

$$\begin{aligned} \dot{W} &= \alpha(\beta - 1)V_a(\sin \psi_t \cos \psi_d - \cos \psi_t \sin \psi_d) \\ \Rightarrow \dot{W} &= \alpha(\beta - 1)V_a \sin(\psi_t - \psi_d) \end{aligned} \quad (52)$$

Based on the UAV initial position, three cases are considered.

Case 1: $\beta < 1$

Using Eqs. (38) and (52),

$$\dot{W} = \alpha(\beta - 1)V_a \sin(\psi_o) \quad (53)$$

Substituting the value of ψ_o from Eq. (46) in Eq. (53) yields

$$\dot{W} = \alpha(\beta - 1)V_a \cos \left(\sin^{-1} \left(\frac{1}{1 + k_e(\beta - 1)^2} \right) \right) \quad (54)$$

As $\alpha > 0$, $V_a > 0$ and because $\sin^{-1}(1/(1 + k_e(\beta - 1)^2)) \in (0, \pi/2)$ for $0 < \beta < 1$ and $\cos \theta > 0$ for all $\theta \in (0, \pi/2)$, Eq. (54) implies

$$\dot{W} < 0 \quad (55)$$

Case 2: $\beta > 1$

Using Eqs. (38) and (52),

$$\dot{W} = -\alpha(\beta - 1)V_a \sin(\psi_o) \quad (56)$$

Using Eqs. (46) and (56),

$$\dot{W} = -\alpha(\beta - 1)V_a \cos \left(\sin^{-1} \left(\frac{1}{1 + k_e(\beta - 1)^2} \right) \right) \quad (57)$$

Note that $\sin^{-1}(1/(1 + k_e(\beta - 1)^2)) \in (0, \pi/2)$ for $1 < \beta < \infty$, and $\cos \theta > 0$ for all $\theta \in (0, \pi/2)$ with $V_a > 0$ and $\alpha > 0$, and hence Eq. (57) implies

$$\dot{W} < 0 \quad (58)$$

Case 3: $\beta = 1$

Using Eqs. (38) and (46),

$$\psi_o = 0 \Rightarrow \psi_d = \psi_t \quad (59)$$

From Eqs. (52) and (59),

$$\dot{W} = 0 \quad (60)$$

Thus, from Eqs. (55), (58), and (60),

$$\dot{W} < 0, \quad \beta \neq 1 \quad \text{and} \quad \dot{W} = 0, \quad \beta = 1 \quad (61)$$

□

Remark 4: Proposition 4 provides the path-following guarantee for the proposed guidance logic, considering that low-level controllers of the UAV accurately track the commanded heading angle ψ_d .

VI. Simulation Results

This section covers numerical simulation studies for the proposed guidance method. Unless specified otherwise, the UAV speed is considered as $V_a = 15$ m/s. Further, $g = 9.8$ m/s² and $k_e = 0.5$ are used. The simulation studies consider a second-order heading hold autopilot [29], which is expressed as

$$\ddot{\psi} = k_\psi(\psi_d - \psi) + k_{\dot{\psi}}(\dot{\psi}_d - \dot{\psi}) \quad (62)$$

where $k_\psi = 100$ and $k_{\dot{\psi}} = 10$ are controller gains, whereas ψ_d denotes the desired heading angle as governed by Eq. (38). Note that the implementation in Eq. (62) requires desired heading rate $\dot{\psi}_d$, which can be computed using Eqs. (38), (43), and (46) as

$$\dot{\psi}_d = \begin{cases} \dot{\psi}_t - \dot{\psi}_o, & 0 < \beta \leq 1 \\ \dot{\psi}_t + \dot{\psi}_o, & \beta \geq 1 \end{cases} \quad (63)$$

Using Eqs. (24), (25), and (43–45), path-tangential rate $\dot{\psi}_t$ can be deduced as

$$\dot{\psi}_t = \frac{F_x^f F_{yy}^f \dot{y} - F_y^f F_{xx}^f \dot{x}}{(F_x^f)^2 + (F_y^f)^2} \quad (64)$$

where

$$F_{xx}^f = \frac{\partial^2 F^f(x, y)}{\partial x^2} = \begin{cases} \frac{n(n-1)x^{n-2}}{a^n}, & x \in (0, a\beta^{1/n}] \\ \frac{n(n-1)(-x)^{n-2}}{a^n}, & x \in [-a\beta^{1/n}, 0) \end{cases} \quad (65)$$

and

$$F_{yy}^f = \frac{\partial^2 F^f(x, y)}{\partial y^2} = \begin{cases} \frac{n(n-1)y^{n-2}}{b^n}, & y \in (0, b\beta^{1/n}] \\ \frac{n(n-1)(-y)^{n-2}}{b^n}, & y \in [-b\beta^{1/n}, 0) \end{cases} \quad (66)$$

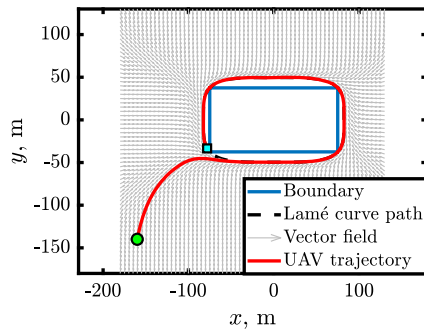
Using Eq. (46), the shaping function rate $\dot{\psi}_o$ is derived as

$$\dot{\psi}_o = \frac{2k_e \dot{\beta}}{(1 + k_e(\beta - 1)^2) \sqrt{k_e(\beta - 1)^2 + 2k_e}} \quad (67)$$

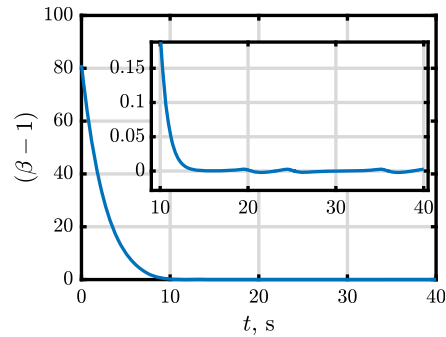
A. Case 1: Sample Scenario

Consider a rectangular boundary of dimensions $(l, m) = (150$ m, 75 m). The bank angle limit for the UAV is considered as $\phi_{\max} = \pm 45$ deg. Using V_a, g and $\phi_{\max}$ in Eq. (35), the corresponding maximum heading-rate and curvature are deduced as $\dot{\psi}_{\max} = 0.6533$ rad/s and $\kappa_{\max_u} = 0.0435$ m⁻¹, respectively. Subsequently, a minimal path-length Lamé curve is obtained using Algorithm 1, with parameter values $(n^*, a^*, b^*) = (4.0631$ m, 82.4549 m, 49.6533 m) and path length 467.53 m. Figure 5a presents the boundary and the optimal Lamé curve path.

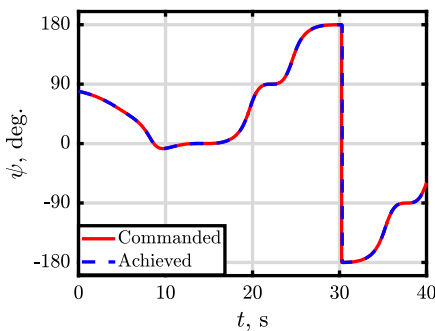
To follow that generated Lamé curve path, consider the initial position of the UAV as $(x_0, y_0) = (-160$ m, -140 m). The initial



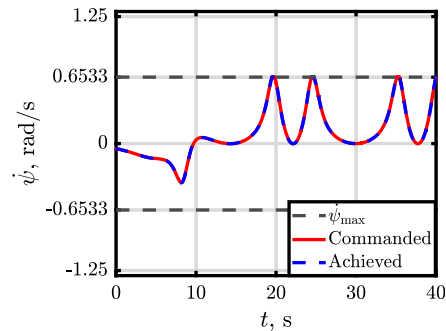
a) Boundary, circumscribing path, and UAV trajectory



b) Tracking error profile



c) UAV heading angle



d) UAV heading-rate

Fig. 5 Results for Case 1: Lamé curve circumscription.

heading angle and heading rate are computed as $\psi(0) = \psi_d(0) = 1.3810$ rad and $\dot{\psi}(0) = \dot{\psi}_d(0) = -0.0498$ rad/s, using Eqs. (38) and (63), respectively. Figure 5 illustrates the simulation results. Figure 5a plots the UAV trajectory. The corresponding tracking error eventually reduces to a negligible value, as depicted in Fig. 5b. The heading angle variation wrapped in $(-180, +180]$ deg is plotted in Fig. 5c. Figure 5d plots the UAV heading-rate profile, wherein the heading rate remains within the vehicle limit $\dot{\psi}_{\max} = 0.6533$ rad/s throughout the course of motion.

Next, a comparison is carried out considering elliptic circumscription of the same rectangular boundary. Using $n = 2$, $(l, m) = (150 \text{ m}, 75 \text{ m})$ in Eq. (26), results in

$$\frac{150^2}{4a^2} + \frac{75^2}{4b^2} = 1 \quad (68)$$

Further, equating vehicle turning-rate limit $\kappa_{\max}$ to the maximum curvature (a/b^2) of the ellipse and using Eq. (68), a unique ellipse with $(a, b) = (111.6797, 50.6109)$ m is obtained. Substituting $(n, a, b) = (2, 111.6797, 50.6109)$ in Eq. (23) deduces the path length of the ellipse to be 528.065 m. Accordingly, Lamé curve circumscription leads to a 11.46% reduction in the path length while utilizing the same maximum turning-rate capability as needed for elliptic circumscription. Figure 6a shows comparative circumscription. Figure 6b shows the UAV heading-rate profiles, wherein the maximum value matches for the Lamé curve and elliptic paths.

B. Case 2: Different UAV Initial Conditions

Further studies are carried out considering UAV initial conditions as listed in Table 1, and the circumscribing Lamé curve path as used in Case 1. Because the UAV maximum turn rate corresponds to the maximum curvature on the Lamé curve path, for UAV initial conditions inside the boundary, the trajectories would typically have a maximum curvature that is higher than that of the Lamé curve path. Accordingly, while utilizing the same bank angle limits, we consider $V_a = 10$ m/s for the UAV initial position inside the boundary. Figure 7 illustrates the simulation results. Figures 7a and 7b show the UAV trajectories for initial positions outside and inside the boundary, respectively. The corresponding tracking error profiles are plotted in Figs. 7c and 7d. The bank angle profiles are shown in Figs. 7e and 7f. The simulation results present an accurate following of the desired Lamé curve path for arbitrary UAV initial positions.

C. Case 3: Sample Scenario with Nonideal Initial Condition

This case considers nonideal initial conditions for the UAV, where the initial heading angle $\psi(0)$ and initial heading rate $\dot{\psi}(0)$ differ from the desired values $\psi_d(0)$ and $\dot{\psi}_d(0)$. The simulations use the same parameters as in Case 1 for (l, m) , $\phi_{\max}$, V_a , g , and $\dot{\psi}_{\max}$. Taking into account the vehicle's limits for $\phi_{\max}$ and $\dot{\psi}_{\max}$, the optimal Lamé curve is generated using Algorithm 1. To follow the optimal Lamé curve, the initial position is set close to the path at $(x_0, y_0) = (82.5549 \text{ m}, 0.1 \text{ m})$. The initial heading angle is 0 rad, which is

Table 1 Lamé curve following with different initial conditions

UAV relative position	Initial positions $(x_0 \text{ m}, y_0 \text{ m})$	Initial heading angle $\psi(0)$ rad [Eq. (38)]	Initial heading rate $\dot{\psi}(0)$ rad/s [Eq. (63)]
Outside the boundary	(140, 50)	-2.8444	-0.0099
—	(-150, 100)	-1.1580	0.1079
—	(-100, -110)	1.4731	-0.0380
—	(90, -120)	1.6220	0.0172
Inside the boundary	(40, 15)	1.1380	0.7378
—	(-15, 20)	2.3688	-0.0030
—	(-25, -5)	-2.3504	0.2880
$(V_a = 10 \text{ m/s})$	(12, -25)	-0.7897	-0.0492

perpendicular to the desired heading along the path, and the initial heading rate is set to 0 rad/s. The guidance gain is selected as $k_e = 0.05$, whereas the controller gains are set to $(k_\psi, k_{\dot{\psi}}) = (4.5, 10)$. Figure 8 presents the simulation results, with the UAV's trajectory shown in Fig. 8a. The corresponding tracking error decreases to a negligible level, as seen in Fig. 8b. Variation of the heading angle within the range $(-180, +180]$ deg is shown in Fig. 8c. Figure 8d depicts the UAV's heading-rate profile, showing that it remains within the allowable limits.

D. Case 4: Different Rectangular Aspect Ratios and Maximum Bank Angle Limits

Consider $\mu = m/l$ as the aspect ratio of the rectangular boundary. Simulations consider $l = 150$ m, and $\mu \in [0.05, 1]$. The maximum bank angle $\phi_{\max}$, $\dot{\psi}_{\max}$, and $\kappa_{\max}$ are identical to the ones used in Case 1. Using Algorithm 1, the optimal Lamé curve path is generated for the choice of μ . Figure 9 illustrates the simulation results. Figure 9a plots the variation in the perimeter of the rectangle, path length of the circumscribing elliptic path, and path length of the optimal Lamé curve path. Percentage reduction in path length gained against elliptical circumscription of the boundary is depicted in Fig. 9b, wherein it can be seen that the proposed Lamé curve paths offer reduced path length in circumscribing rectangular boundaries for all choices of μ . The reduction in path length is more than 10% for $\mu > 0.476$, and it is more than 30% for $\mu > 0.751$. Further, studies are carried out considering $l = 100$ m and $l = 200$ m. The overall results are plotted in Fig. 9c. The path-length advantage increases with an increase in choice of l . This is attributed to the elongation of the circumscribing elliptic path along the length of the rectangular boundary while satisfying the same turn-rate requirement. Additionally, simulations are carried out considering the rectangular boundary as used in Case 1, and a variable maximum bank angle limit as illustrated in Fig. 10. Figure 10a plots the percentage reduction in path length with variation in the UAV bank angle limit. Figure 10b shows the percentage reduction in path length for different choices of l while varying the UAV bank angle limit within the same range as

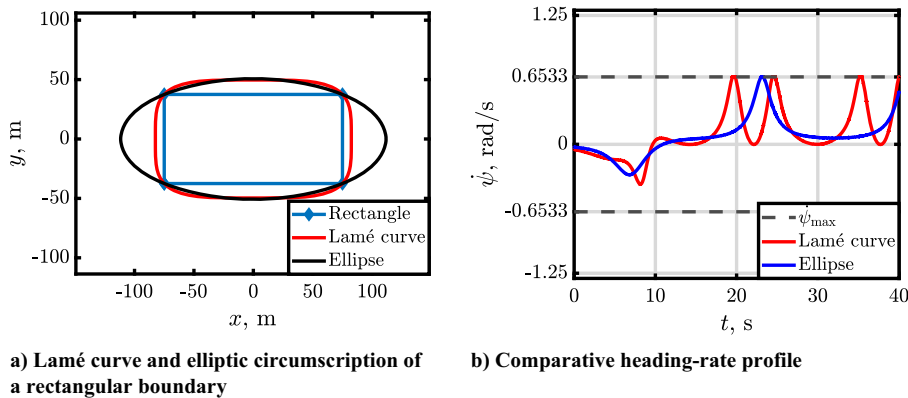


Fig. 6 Results for Case 1: Comparison with elliptic circumscription.

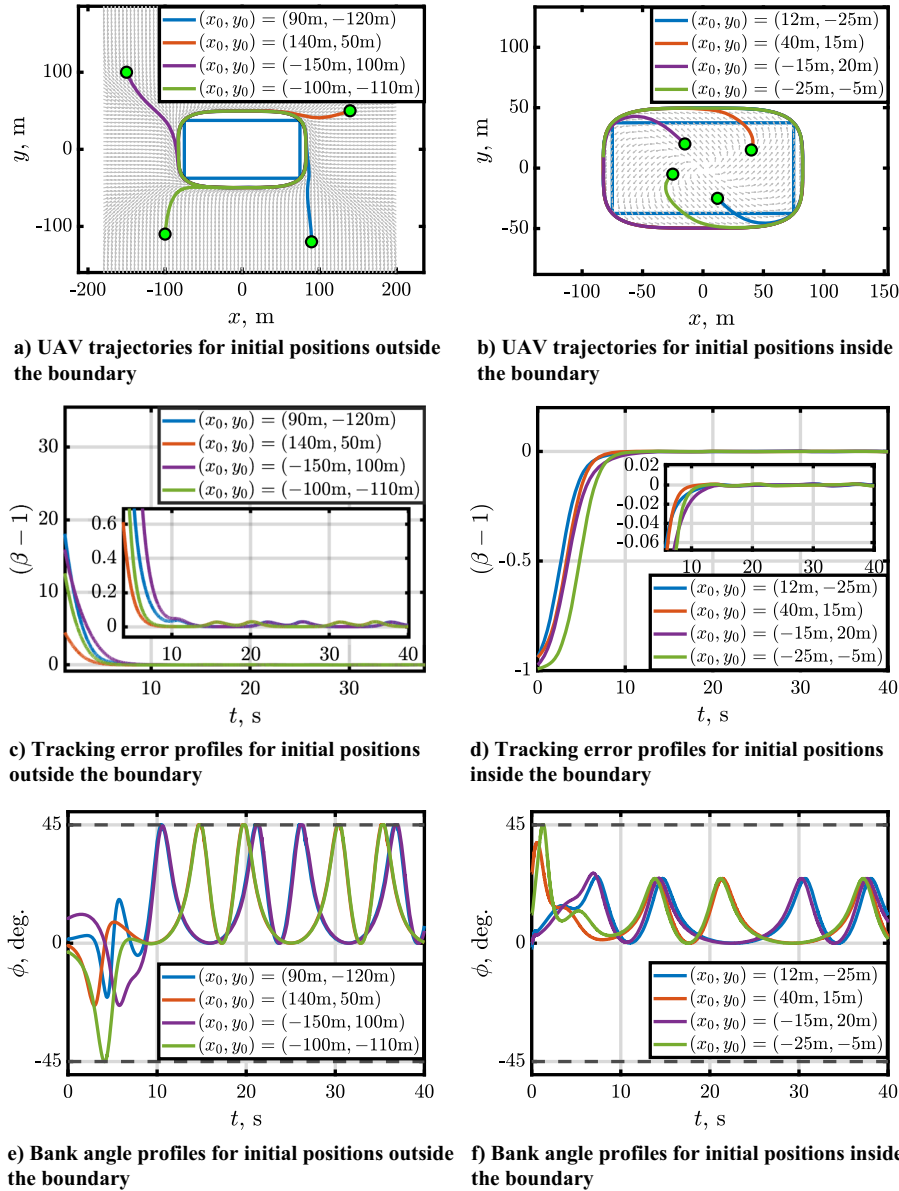


Fig. 7 Results for Case 2: Different UAV initial conditions.

used in Fig. 10a. Again, results highlight significant path-length advantage in using Lamé curve circumscription as compared to the commonly used elliptic paths.

VII. Experimental Validation Results

This section discusses experimental validation results for the proposed guidance method. Experimental studies utilize the Crazy-Flie 2.0 quadrotor platform and an indoor motion capture facility equipped with OptiTrack Prime-13 cameras. The quadrotor is commanded to fly with a prescribed speed with pitch and roll angle limits, thereby mimicking a fixed-wing aircraft motion. A brief description of the experimental setup and the key subsystems is discussed subsequently.

A. Experimental Setup

The experimental setup comprises the Motion Capture (MoCap) system, command and communication systems, and the Crazyflie 2.0 quadrotor platform equipped with infrared markers. Figure 11 illustrates the exchange of information between the components, and communication methods utilized within the experimental framework. The MoCap system, comprising cameras and software, records the movement of the quadrotor. The LED illumination ring around

each of the cameras emits infrared light, which is reflected by markers placed on the quadrotor and captured by the camera lens. The processed data from the camera is transmitted to Optitrack Motive software of the MoCap system through an ethernet protocol, facilitating an accurate determination of the position and orientation of the quadrotor.

The command center uses MATLAB Simulink operating on a Windows PC and generates reference position and velocity commands for the autopilot control loops to execute the path-following guidance strategy. The communication center runs on a Linux PC, utilizing the Robotic Operating System to establish communication links with the Crazyflie 2.0, MoCap, and the command center, as illustrated in Fig. 12. Further, the communication between the Crazyflie 2.0 and the command and communication center is facilitated at a relay rate of 100 Hz through a long-range open-source USB bidirectional radio dongle called Crazyradio. The experiments utilize Crazyflie 2.0 nano quadrotor, which is a widely used open-source UAV platform [30–33]. The hardware and software architecture details of the quadrotor can be found in Ref. [34].

B. Guidance and Control Architecture

As indicated in Fig. 12, the proposed guidance methods are implemented in “Vector-field guidance block” subsystem, to generate the

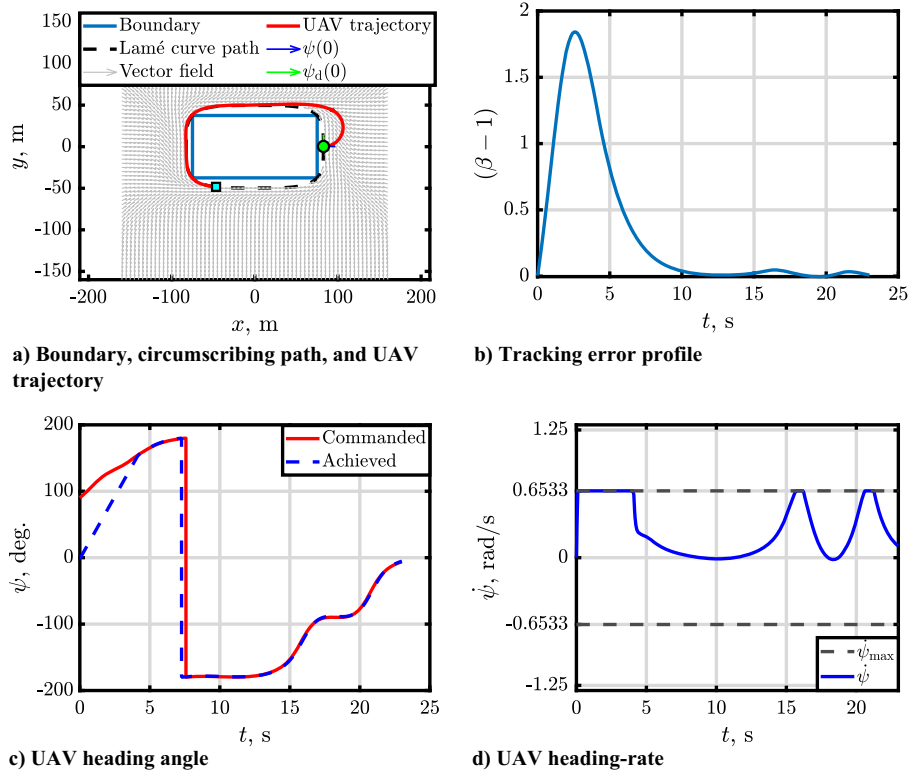


Fig. 8 Results for Case 3: Following Lamé curve path with nonideal initial condition.

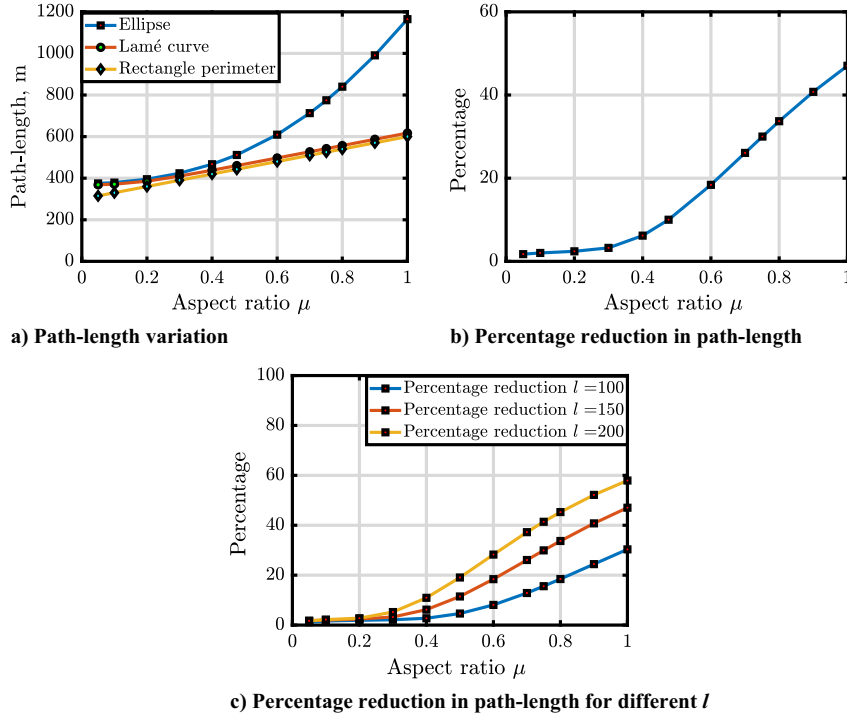


Fig. 9 Results for Case 4: Path-length comparison with variation in rectangular aspect ratios for $\phi_{\max} = \pm 45^\circ$.

reference position and velocity commands. The reference velocity commands $\dot{x}_{\text{ref}}, \dot{y}_{\text{ref}}$ are obtained by substituting the proposed guidance command ψ_d from Eq. (38) into Eqs. (24) and (25) as

$$\dot{x}_{\text{ref}}(t) = V_{\text{ref}} \cos \psi_d \quad (69)$$

$$\dot{y}_{\text{ref}}(t) = V_{\text{ref}} \sin \psi_d \quad (70)$$

Accordingly, reference positions $x_{\text{ref}}(t), y_{\text{ref}}(t)$ are obtained using Eqs. (69) and (70) as

$$x_{\text{ref}}(t) = x(0) + \int_0^t \dot{x}_{\text{ref}}(t) dt \quad (71)$$

$$y_{\text{ref}}(t) = y(0) + \int_0^t \dot{y}_{\text{ref}}(t) dt \quad (72)$$

As shown in Fig. 12, quadrotor states are controlled using “Autopilot dynamics” subsystem for executing the reference guidance commands. Crazyflie 2.0 quadrotor operates with four distinct control inputs, namely, total thrust u_1 , commanded roll ϕ_c , commanded pitch θ_c , and commanded yaw rate $\dot{\psi}_{qc}$:

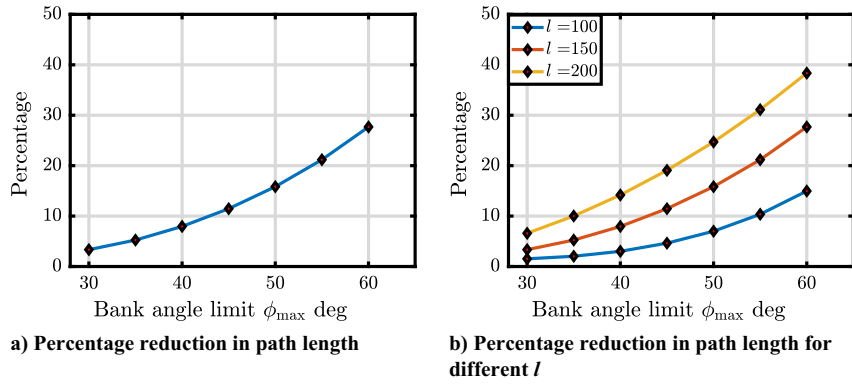


Fig. 10 Results for Case 4: Path-length comparison with variation in maximum bank angle limit for $\mu = 0.5$.

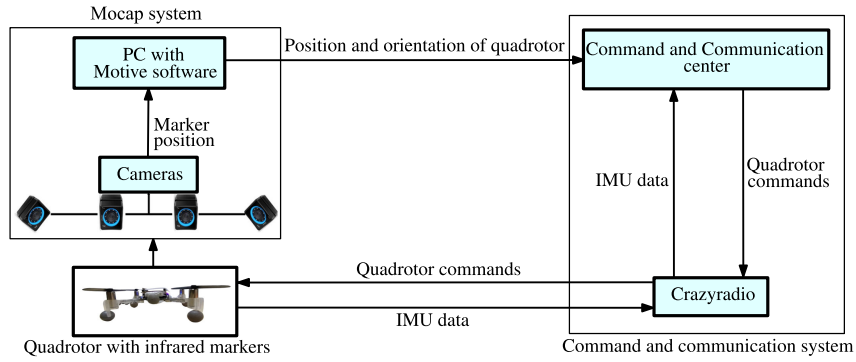


Fig. 11 Overview of system architecture.

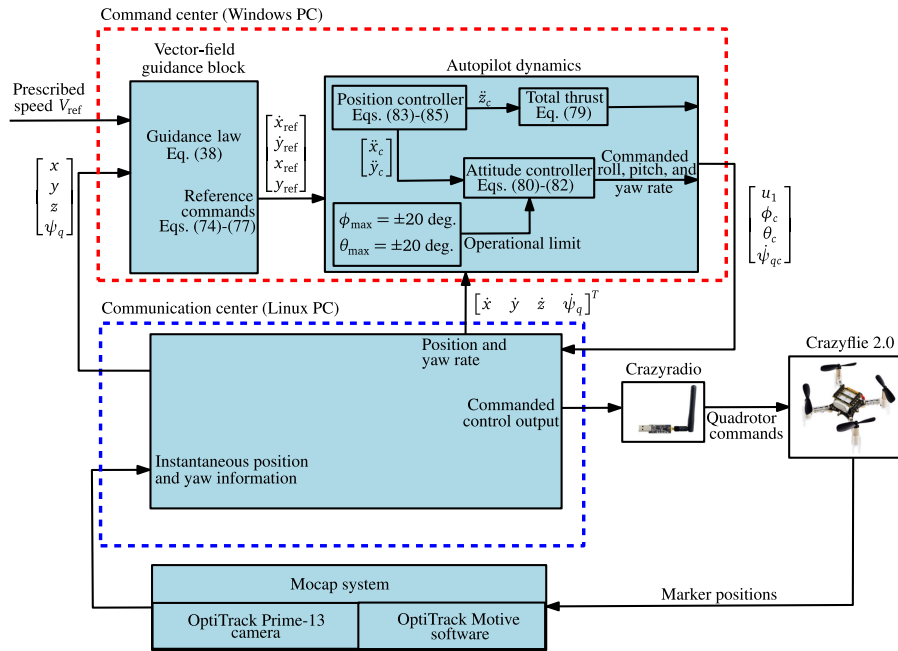


Fig. 12 Overall guidance, and control architecture.

$$u = [u_1 \quad \phi_c \quad \theta_c \quad \dot{\psi}_{qc}]^T \quad (73)$$

$$\phi_c = \frac{1}{g} (\ddot{x}_c \sin \psi_{qc} - \ddot{y}_c \cos \psi_{qc}) \quad (75)$$

The total thrust and commanded controlled inputs are related to the reference position and velocity commands as [35]

$$u_1 = M(g + \ddot{z}_c) \quad (74)$$

$$\theta_c = \frac{1}{g} (\ddot{x}_c \cos \psi_{qc} + \ddot{y}_c \sin \psi_{qc}) \quad (76)$$

$$\dot{\psi}_{qc} = 0 \quad (77)$$

where M and g denote the mass of Crazyflie 2.0, and acceleration due to gravity, respectively. The autopilot generates these control inputs for the quadrotor to track the reference commands. These control inputs are transmitted to the quadrotor through the Crazyradio dongle. Further, the commanded translational acceleration $\ddot{x}_c, \ddot{y}_c, \ddot{z}_c$ are related to the reference position and velocity commands through a proportional-derivative control logic as [36]

$$\ddot{x}_c = k_{px}(x_{\text{ref}} - x) + k_{dx}(\dot{x}_{\text{ref}} - \dot{x}) \quad (78)$$

$$\ddot{y}_c = k_{py}(y_{\text{ref}} - y) + k_{dy}(\dot{y}_{\text{ref}} - \dot{y}) \quad (79)$$

$$\ddot{z}_c = k_{pz}(z_{\text{ref}} - z) + k_{dz}(\dot{z}_{\text{ref}} - \dot{z}) \quad (80)$$

where k_{px}, k_{py}, k_{pz} and k_{dx}, k_{dy}, k_{dz} are tunable gains, and the reference altitude and altitude rate are denoted as z_{ref} and $\dot{z}_{\text{ref}}$, respectively.

C. Experimental Results

The flight trials consider initial position of the Crazyflie 2.0 as $(x(0), y(0), z(0)) = (-1.5 \text{ m}, 1.5 \text{ m}, -1 \text{ m})$. The prescribed speed is considered as $V_{\text{ref}} = 0.25 \text{ m/s}$. Because the experimental studies consider a planar path-following scenario, the reference altitude is considered as $z_{\text{ref}} = -1 \text{ m}$ (z axis points downward with respect to a ground plane) and the reference altitude rate is set as $\dot{z}_{\text{ref}} = 0 \text{ m/s}$. Further, $\psi_c = 0$ is considered throughout the course of motion. The following constants are used in the studies: $(k_{px}, k_{py}, k_{pz}, k_{dx}, k_{dy}, k_{dz}, k_e) = (4.05, 5.6, 7, 3.5, 4.7, 5.1, 0.1)$. The first experiment considers $(n, a, b) = (2.5, 1 \text{ m}, 0.75 \text{ m})$ and Fig. 13 illustrates the experimental results. Figure 13a shows the Crazyflie 2.0 trajectory, and Fig. 13b plots the corresponding tracking error profile showing negligible errors. The tracking of reference position commands is shown in Figs. 13c and 13d. Further, experiments are carried out considering $n = \{3, 3.5, 4, 4.5\}$ with the choice of (a, b) , as used in the first experiment. The

Crazyflie 2.0 trajectory plots are shown in Figs. 14a–14d, where the quadrotor follows the desired path, and the corresponding tracking error profiles reducing to a very low value are plotted in Fig. 14e. The comprehensive experimental investigation of the Lamé curve path-following is presented for the first time in the literature, and the results demonstrate the efficacy of the proposed vector-field guidance method.

VIII. Conclusions

Utilizing Lamé curve paths, this paper introduces a new guidance method for rectangular boundary surveillance. Curvature and path-length properties of the Lamé curve paths are analyzed. Subject to maximum curvature constraint, an algorithm deduces the minimum path-length Lamé curve for circumscribing a rectangular boundary. As a main contribution, it is shown that, while satisfying the same maximum unmanned aerial vehicle (UAV) turn rate, circumscribing a rectangular boundary using the Lamé curve path offers a significant reduction in path length compared to the widely used elliptic circumscription. The advantage is shown to enhance with increasing aspect ratio and dimensions of the rectangle. Further, a vector-field guidance method governing the UAV heading angle is proposed to follow the circumscribing Lamé curve path. A simple and closed-form structure of the commanded heading-angle encapsulating path tangential term and arcsine shaping function facilitates smooth variation of the UAV heading angle for following the desired path. Simulation studies are carried out by considering a second-order heading-hold autopilot and different UAV initial conditions. Results show accurate following of the desired path from arbitrary UAV initial conditions, highlighting the effectiveness of the proposed guidance method. Comparative studies considering variation in rectangular boundary dimensions and UAV bank angle limits demonstrate the superiority of the proposed method over elliptic surveillance. Furthermore, experimental validation studies carried out in an indoor motion-capture facility while utilizing the Crazyflie 2.0 quadrotor UAV platform establish the practical applicability of the proposed guidance method. The proposed vector-field guidance method uses a heading-based guidance input, which makes the method readily applicable to a variety of autonomous vehicles like quadrotor UAVs, and ground

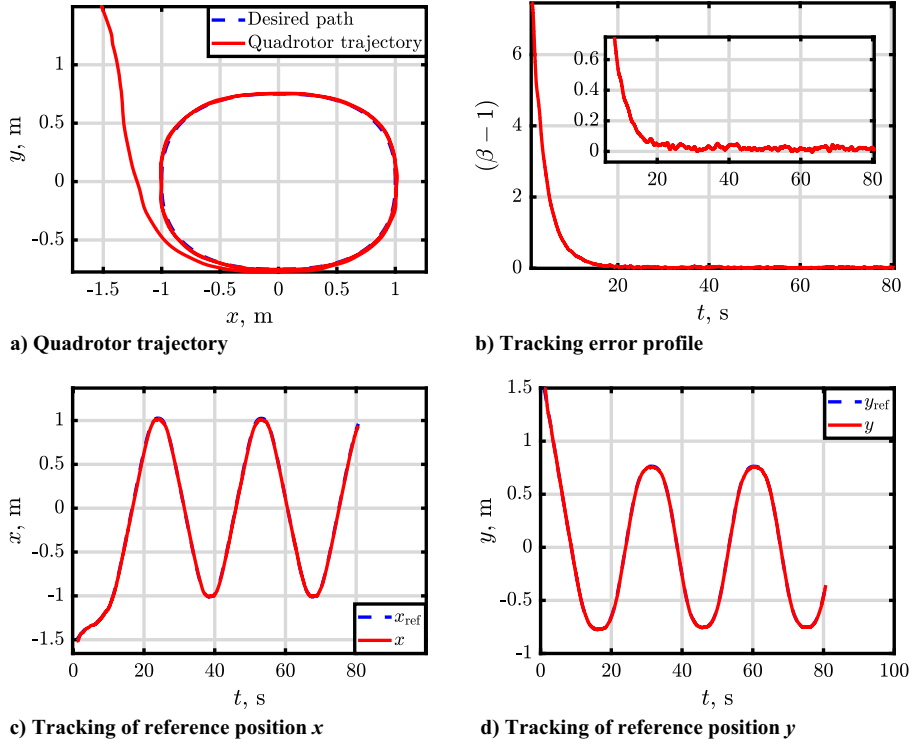


Fig. 13 Experimental results for following a Lamé curve path with parameters $(n, a, b) = (2.5, 1 \text{ m}, 0.75 \text{ m})$.

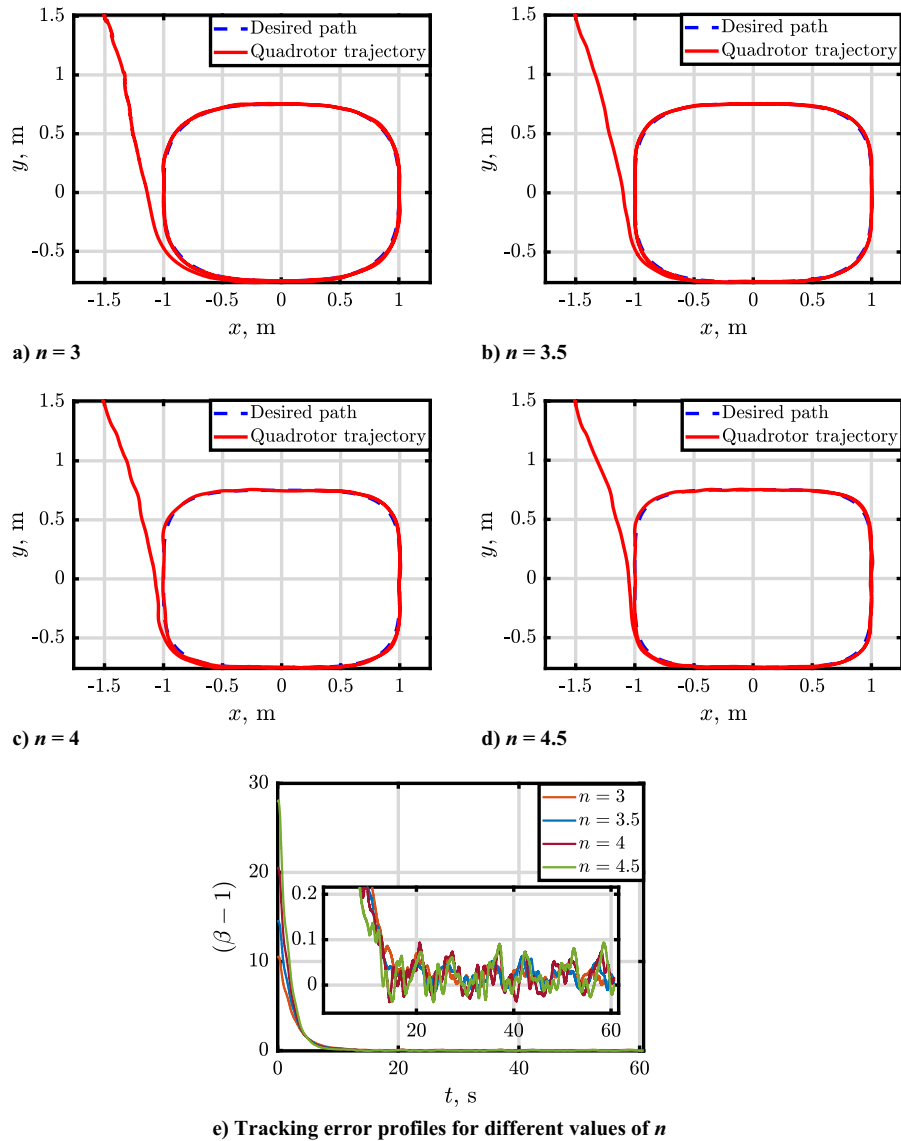


Fig. 14 Experimental results showing quadrotor trajectories and tracking error profiles for different values of n .

and underwater robots. Future research directions include further developing the method to three-dimensional scenarios, and validation studies considering the presence of wind.

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Appendix: Proof of Lemma 1

Consider the instantaneous UAV position lies in the first quadrant. Accordingly, for $x \in [0, a\beta^{1/n}]$, $y \in (0, b\beta^{1/n}]$, Eq. (39) implies

$$\frac{x^n}{a^n\beta} + \frac{y^n}{b^n\beta} = 1 \quad (A1)$$

Using Eq. (A1), the slope at Q can be computed as

$$\tan \psi_t = \left. \frac{dy}{dx} \right|_{(x_u, y_u)} = -\left(\frac{b}{a}\right)^n \left(\frac{x_u}{y_u}\right)^{n-1} \quad (A2)$$

A line joining origin and point Q intersects the desired Lamé curve at the point $P(x_p, y_p)$, as shown in Fig. 4. Hence, the equation of the line can be expressed as

$$y = \frac{y_u}{x_u} x \quad (A3)$$

Solving Eq. (3) and Eq. (A3), the point P can be deduced as

$$x_p = \frac{1}{\sqrt[n]{\frac{1}{a^n} + \left(\frac{y_u}{x_u}\right)^n \frac{1}{b^n}}}, \quad y_p = \frac{y_u}{x_u} \frac{1}{\sqrt[n]{\frac{1}{a^n} + \left(\frac{y_u}{x_u}\right)^n \frac{1}{b^n}}} \quad (A4)$$

Finally, using Eq. (3) and Eq. (A4), the slope on the desired Lamé curve at the point $P(x_p, y_p)$ is deduced as

$$\left. \frac{dy}{dx} \right|_{(x_p, y_p)} = -\left(\frac{b}{a}\right)^n \left(\frac{x_p}{y_p}\right)^{(n-1)} = -\left(\frac{b}{a}\right)^n \left(\frac{x_u}{y_u}\right)^{(n-1)} \quad (A5)$$

From Eq. (A2) and Eq. (A5), it can be concluded that for all UAV positions in the first quadrant, the slope at the point Q on the concentric Lamé curve is equal to the slope at the point P on the desired Lamé curve. Similar analysis can be carried out to prove the property in the remaining three quadrants.

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